RV College of Engineering®
Mysore Road, RV Vidyaniketan Post
Bengaluru - 560059, Karnataka, India

Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF INFORMATION SCIENCE & ENGINEERING
(Common to CS, IS & AI)

Date: 14/1/2026	IMPROVEMENT CIE	Max. Marks: 60
Semester: V	UG	Duration: 2 Hour
Course Title: ADVANCED ALGORITHMS		Course Code: CS355TBB

Sl. No.	PART A	M	BT	CO
1.	In modular arithmetic, a≡b(mod n)a means _____	1	1	3
2.	The worst-case time complexity of naïve string matching is _____	1	2	2
3.	Write two limitations of Fibonacci heaps.	2	2	3
4.	_____string matching algorithm guarantees linear time in all cases	1	1	4
5.	Under which condition does Rabin–Karp algorithm degrade to O(nm)?	1	2	4
6.	The time complexity of Euclid's algorithm for inputs a and b can be given as _____	2	2	3
7.	For pattern "ABAB", calculate the LPS value of the last character?	1	3	2
8.	____ is the time complexity of merging two Fibonacci heaps.	1	3	2
	PART B			
1a.	Write two possible solutions for the modular equation 35x≡10(mod50).	5	3	3
1b.	Explain the RSA cryptosystem.	5	1	2
2.	With the help of an example, Illustrate the DECREASE-KEY operation in a Fibonacci heap using a neat diagram. Show how cascading cuts are performed and explain their significance.	10	3	4
3.	Construct a binomial queue by inserting the following keys in order: 12, 7, 25, 15, 28, 33, 41. Show all intermediate binomial trees formed during insertion.	10	3	2
4.	Apply the Chinese Remainder Theorem to solve a system of simultaneous congruences: x≡2(mod3), x≡3(mod4) and x≡1(mod5). Show all intermediate steps and verify the solution.	10	3	1
5.	Find all occurrences of the pattern P="ABABCABAB" in the text T="ABABDABACDABABCABAB", using Knuth–Morris–Pratt (KMP) algorithm. Explain the steps and Give the analysis of the algorithm.	10	3	3